

Unified Origin of the Radial Acceleration Relation and the Hubble Tension via Entropic Gravity Modification

EFC- H_0 Unification Paper v1.0

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Abstract

We demonstrate that the Hubble tension (H_0 discrepancy between early and late universe measurements) and the radial acceleration relation (RAR) in galaxies arise from the same physical mechanism: an entropy-driven modification of effective gravity. Using the Energy-Flow Cosmology (EFC) Core Lock framework with sigmoid entropy mapping $S(z)$, we derive the gravitational coupling coefficient $a_G \approx 0.094$ from Hubble tension data. Independently, Modified Newtonian Dynamics (MOND) phenomenology at the transition regime ($g_{\text{bar}}/a_0 \approx 5$) implies $a_G \approx 0.107$. The 14% agreement, combined with the Friedmann-derived constraint $a_{H_0} = \frac{1}{2}a_G$, enables prediction of $H_0 \approx 74$ km/s/Mpc from galaxy rotation data alone—matching the observed value of 73.0 km/s/Mpc within 1.2%.

Keywords: Hubble tension, radial acceleration relation, MOND, entropic gravity, modified gravity, Energy-Flow Cosmology

1 Introduction

Two of the most significant anomalies in modern astrophysics—the Hubble tension and the mass discrepancy in galaxies—are conventionally treated as separate problems requiring independent solutions. The Hubble tension refers to the 4–6 σ discrepancy between H_0 measured from the cosmic microwave background (CMB, ~ 67.4 km/s/Mpc) versus local distance ladder methods (~ 73 km/s/Mpc) (3; 4). The radial acceleration relation (RAR), quantified in the SPARC database (5; 2), shows a tight correlation between observed and baryonic accelerations in galaxies that standard Λ CDM struggles to explain without fine-tuning.

In this paper, we propose that both phenomena arise from a single mechanism: entropy-driven modification of effective gravitational coupling. The framework is based on Energy-Flow Cosmology (EFC), specifically the Core Lock dynamical engine (1), which posits that physical observables project via an accumulated entropy phase $\Phi(S)$.

The central claim is that the gravitational coupling coefficient $a_G \approx 0.1$, derived from MOND/RAR phenomenology in the acceleration transition zone, when combined with the Friedmann equation constraint $a_{H_0} = \frac{1}{2}a_G$, predicts the observed H_0 tension without free parameters.

2 Theoretical Framework

2.1 Core Lock Dynamics

The EFC Core Lock mechanism defines a running parameter $f(S)$ with constant negative beta function along the entropy axis S . The unique solution satisfying the beta-constraint $\beta_f = d \ln f / dS = -1/\Delta S$ is:

$$f(S) = f_0 \exp\left(-\frac{S}{\Delta S}\right) \quad (1)$$

Physical observables project via the accumulated phase:

$$\Phi(S) = \int_0^S \frac{dS'}{f(S')} = \frac{\Delta S}{f_0} \left[\exp\left(\frac{S}{\Delta S}\right) - 1 \right] \quad (2)$$

With normalization $\Delta S = f_0 = 1$, this simplifies to $\Phi(S) = e^S - 1$. The projection law for observable X is:

$$\ln\left(\frac{X}{X_0}\right) = a_X \cdot \Phi(S) \quad (3)$$

where a_X is the coupling coefficient for parameter X .

2.2 Entropy Mapping $S(z)$

We define the entropy parameter as a sigmoid function of redshift, motivated by structure formation dynamics:

$$S(z) = \frac{1}{2} \left[1 - \tanh\left(\frac{\ln(1+z) - \ln(1+z_{\text{trans}})}{\Delta}\right) \right] \quad (4)$$

With $z_{\text{trans}} = 3$ (corresponding to the peak of cosmic star formation rate, ‘‘Cosmic Noon’’ (7)) and $\Delta = 0.5$, this yields $S(z=0) \approx 0.996$ and $S(z=1100) \approx 0$. The resulting phase difference is:

$$\Delta\Phi = \Phi(S_{\text{today}}) - \Phi(S_{\text{CMB}}) \approx 1.71 \quad (5)$$

2.3 The Friedmann Constraint

From the Friedmann equation $H^2 \propto G_{\text{eff}} \cdot \rho$, assuming the entropy coupling modifies only G_{eff} (not ρ_{eff}), we have:

$$G_{\text{eff}}(S) = G_0 \exp(a_G \cdot \Delta\Phi) \quad (6)$$

Taking logarithms and differentiating: $2\Delta H/H \approx \Delta G/G$. This yields the critical relation:

$$\boxed{a_{H_0} = \frac{1}{2} a_G} \quad (7)$$

This factor of $\frac{1}{2}$ is not fitted—it emerges directly from the Friedmann equation structure.

3 Connection to MOND/RAR

The MOND paradigm (6) introduces a universal acceleration scale $a_0 = 1.2 \times 10^{-10} \text{ m/s}^2$. The ‘‘simple’’ interpolation function $\mu(x) = x/(1+x)$, where $x = g_{\text{bar}}/a_0$, connects Newtonian and deep-MOND regimes.

In EFC language, the effective gravitational enhancement μ^{-1} maps to the coupling coefficient via:

$$a_G = \frac{\ln(\mu^{-1})}{\Delta\Phi} \quad (8)$$

The highlighted row ($g_{\text{bar}}/a_0 = 5$) represents the transition zone where cosmic-scale structures typically reside.

Table 1: MOND interpolation values and implied a_G

g_{bar}/a_0	μ	μ^{-1}	$\ln(\mu^{-1})$	Implied a_G
1.0	0.500	2.000	0.693	0.406
2.0	0.667	1.500	0.405	0.238
yellow!20 5.0	0.833	1.200	0.182	0.107
10.0	0.909	1.100	0.095	0.056

4 Results

4.1 H_0 Prediction from Hubble Tension Data

From the observed H_0 values (SH0ES: 73.0 km/s/Mpc; Planck: 67.4 km/s/Mpc), we derive:

$$\ln(73.0/67.4) = 0.0798 \quad (9)$$

$$a_{H_0} = 0.0798/1.71 = 0.0467 \quad (10)$$

$$a_G = 2 \times a_{H_0} = 0.0935 \approx 0.094 \quad (11)$$

4.2 a_G from MOND Transition Zone

At the transition regime ($g_{\text{bar}}/a_0 \approx 5$), the MOND interpolation function gives $\mu^{-1} = 1.20$, implying:

$$a_G^{(\text{MOND})} = \frac{\ln(1.20)}{1.71} = 0.107 \quad (12)$$

4.3 Comparison and Prediction

Table 2: Comparison of a_G from independent sources

Source	a_G value	Method
H_0 tension (Phase 1.1)	0.094	Cosmological
MOND transition ($g/a_0 = 5$)	0.107	Galactic

Discrepancy: $|0.107 - 0.094|/0.094 = 14\%$

Using $a_G = 0.107$ from MOND data (independent of H_0), we predict:

$$H_0^{(\text{pred})} = 67.4 \times \exp\left(\frac{0.107}{2} \times 1.71\right) = 67.4 \times 1.096 = \mathbf{73.9 \text{ km/s/Mpc}} \quad (13)$$

Observed: 73.0 km/s/Mpc. **Deviation:** 1.2%

5 Discussion

The agreement between a_G derived from cosmological (H_0 tension) and galactic (MOND/RAR) data suggests a unified origin for both phenomena. The physical interpretation is that the cosmic web and large-scale filamentary structures reside in the MOND transition zone ($g_{\text{bar}} \approx \text{few} \times a_0$), where gravitational enhancement is approximately 10–20%.

Key strengths of this approach:

1. The factor of $\frac{1}{2}$ in $a_{H_0} = \frac{1}{2}a_G$ emerges from Friedmann, not fitting
2. $a_G \approx 0.1$ is constrained by galactic data independent of H_0 measurements

3. The prediction $H_0 \approx 74$ km/s/Mpc matches observation within 1.2%
4. The framework is falsifiable: a_G from detailed SPARC fits must yield ~ 0.1

Limitations and future work: The choice of $g_{\text{bar}}/a_0 = 5$ as the “cosmological average” requires theoretical justification from cosmic web statistics. The sigmoid parameters z_{trans} and Δ should be calibrated from $\sigma_8(z)$ evolution data. A full SPARC rotation curve fit to derive a_G directly would strengthen the claim.

6 Conclusion

We have demonstrated that the H_0 tension and the radial acceleration relation can be understood as manifestations of the same underlying physics: entropy-driven modification of effective gravitational coupling. The key relations are:

$$G_{\text{eff}} = G_0 \exp(a_G \cdot \Delta\Phi(S)) \quad (14)$$

$$a_{H_0} = \frac{1}{2}a_G \quad [\text{from Friedmann}] \quad (15)$$

$$a_G \approx 0.1 \quad [\text{from MOND transition zone}] \quad (16)$$

This yields a prediction of $H_0 \approx 74$ km/s/Mpc from galactic dynamics, matching the observed tension. The EFC framework provides a unified mechanism linking galactic and cosmological scales through a single coupling coefficient $a_G \approx 0.1$.

Epistemic status: The value $a_G \approx 0.1$ inferred from MOND phenomenology should be regarded as an order-of-magnitude consistency check, not yet a direct fit to SPARC rotation curve data. A full empirical determination of a_G from galaxy data constitutes the next step toward definitive confirmation or falsification.

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Version History

Version	Date	Changes
1.0	February 2026	Initial publication